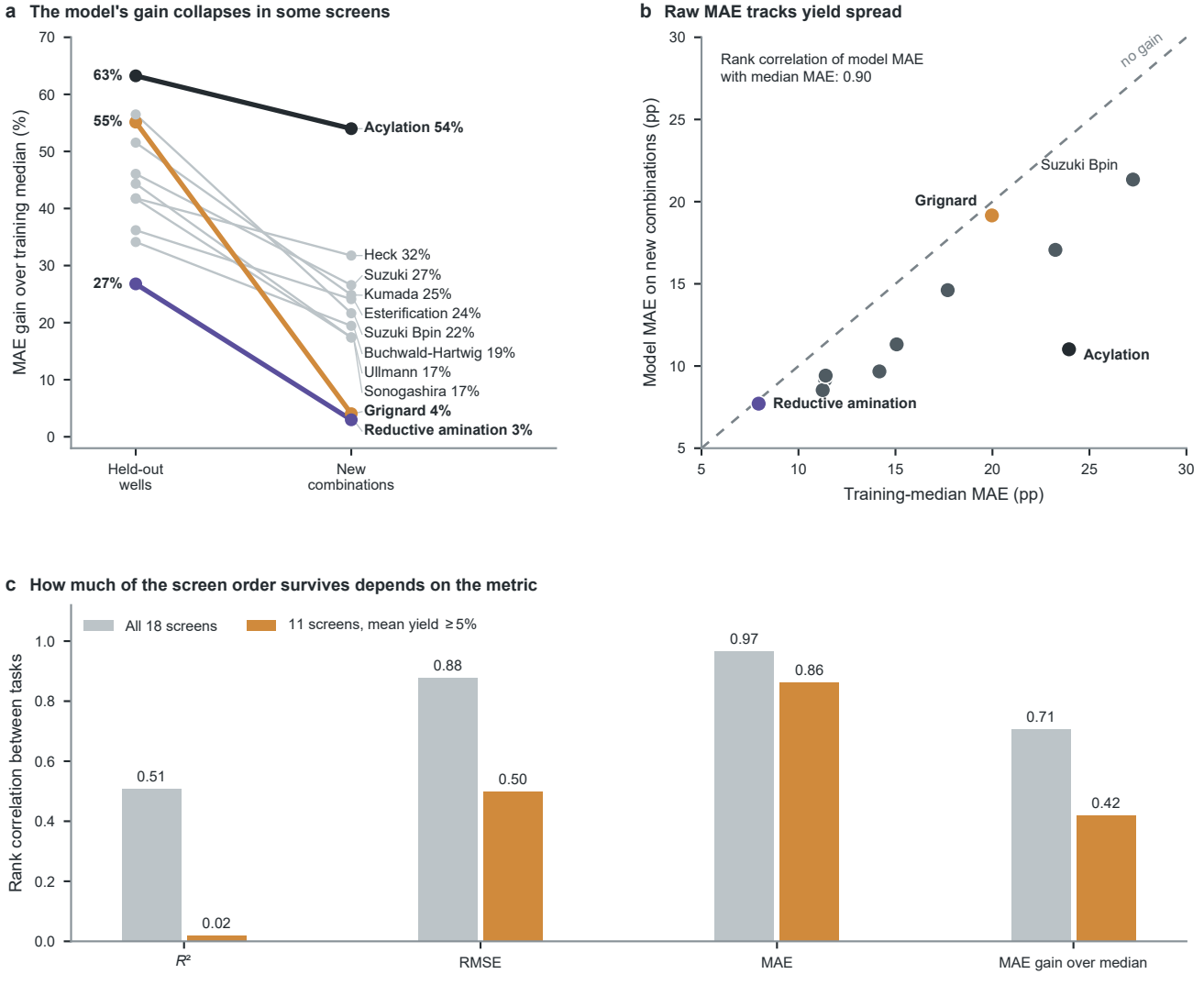

a The model's gain collapses in some screens
MAE gain over training median (%)
70
60
50
40
30
20
10
0
63%
55%
27%
Acylation 54%
Heck 32%
Suzuki 27%
Kumada 25%
Esterification 24%
Suzuki Bpin 22%
Buchwald-Hartwig 19%
Ullmann 17%
Sonogashira 17%
Grignard 4%
Reductive amination 3%
Held-out wells
New combinations
b Raw MAE tracks yield spread
Rank correlation of model MAE with median MAE: 0.90
no gain
Suzuki Bpin
Grignard
Acylation
Reductive amination
Model MAE on new combinations (pp)
Training-median MAE (pp)
30
25
20
15
10
5
c How much of the screen order survives depends on the metric
All 18 screens
11 screens, mean yield ≥5%
Rank correlation between tasks
1.0
0.8
0.6
0.4
0.2
0.0
0.51
0.02
0.88
0.50
0.97
0.86
0.71
0.42
R²
RMSE
MAE
MAE gain over median